

Instructions — How to Reproduce This Project

This document walks through every step needed to set up the environment, download the data, and run the full intrusion detection pipeline from scratch.

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1. System Requirements

Before starting, confirm your machine meets these requirements:

Item	Minimum	Recommended
Operating System	Windows 10, macOS 11, or Ubuntu 20.04	Windows 11 / macOS 13 / Ubuntu 22.04
Python	3.12	3.12
RAM	8 GB	16 GB
Disk Space	2 GB free	5 GB free
GPU	Not required	Not required
Internet	Required (for setup)	—

2. Clone the Repository

Open a terminal (Command Prompt, PowerShell, or Bash) and run:

```
bash

git clone https://github.com/<your-username>/edge-to-cloud-ids.git
cd edge-to-cloud-ids
```

If you do not have Git installed, download it from <https://git-scm.com> first.

Alternatively, click **Code** → **Download ZIP** on GitHub and unzip it.

3. Set Up Python Environment

Using a virtual environment keeps this project's dependencies isolated from other Python projects on your machine. This step is strongly recommended.

Windows:

```
bash

python -m venv venv
venv\Scripts\activate
```

macOS / Linux:

```
bash

python3 -m venv venv
source venv/bin/activate
```

You should see `(venv)` appear at the start of your terminal prompt, confirming the environment is active.

┃ To deactivate the environment later, simply run `deactivate`.

4. Install Dependencies

With the virtual environment active, install all required packages:

```
bash
```

```
pip install -r requirements.txt
```

This installs the following libraries:

Library	Purpose
pandas	Data loading and manipulation
numpy	Numerical operations
matplotlib	Static plots and charts
seaborn	Advanced correlation heatmaps
scikit-learn	All three ML models + preprocessing + metrics
plotly	Interactive visualizations
jupyter	Notebook environment

Installation typically takes 2-5 minutes depending on your internet speed.

5. Download and Place the Dataset

The CICIoT2023 dataset is not included in this repository due to its size. Follow these steps:

Step 1: Go to the official dataset page:

<https://www.unb.ca/cic/datasets/iotdataset-2023.html>

Step 2: Download the training CSV file.

Step 3: Inside the project folder, create this directory structure and place the file there:

```
edge-to-cloud-ids/  
├── CICIOT23/  
    ├── train/  
        └── train.csv    ← place it here
```

Windows (Command Prompt):

```
cmd
```

```
mkdir CICIOT23\train
move train.csv CICIOT23\train\train.csv
```

macOS / Linux:

```
bash

mkdir -p CICIOT23/train
mv ~/Downloads/train.csv CICIOT23/train/train.csv
```

Verify the path is correct by running:

```
bash

# macOS / Linux
ls CICIOT23/train/

# Windows
dir CICIOT23\train\
```

You should see `train.csv` listed.

6. Launch the Notebook

With your virtual environment still active, start Jupyter:

```
bash

jupyter notebook Edge.ipynb
```

Your browser will open automatically. If it does not, copy the URL shown in the terminal (it starts with `http://localhost:8888/...`) and paste it into your browser.

Alternative — VS Code:

1. Open the project folder in VS Code
 2. Install the **Jupyter** extension if prompted
 3. Open `Edge.ipynb`
 4. Select the `venv` interpreter in the top-right kernel picker
 5. Click **Run All** or run cells individually
-

7. Running Each Section

Run the notebook cells **from top to bottom**. The table below describes what each section does.

Section	What It Does	Expected Output
1. Library Imports	Loads all required Python packages	<code>Libraries loaded successfully</code> printed
2. Data Loading	Reads 100,000 rows from <code>train.csv</code>	Dataset shape printed, first 5 rows displayed
3. Class Distribution	Plots the distribution of attack labels	Bar chart of attack type counts
4. Feature Histograms	Plots distributions of the first 5 features	Grid of histograms
5. Correlation Heatmap	Computes feature correlations on a 5,000-row sample	Heatmap grid
6. Missing Value Analysis	Checks for nulls across all 47 features	Bar chart + total missing count
7. Preprocessing	Label encoding + StandardScaler normalization	Before/after feature distribution plots
8. Train-Test Split	80% train / 20% test split	Shape printed + bar chart showing split sizes
9. Logistic Regression	Trains LR, computes metrics, plots results	Accuracy, Precision, Recall, F1 printed + bar chart
10. Random Forest	Trains RF (100 trees), computes metrics	Accuracy, Precision, Recall, F1 printed + bar chart
11. LR vs RF Comparison	Side-by-side accuracy and F1 line plots	Two comparison line charts
12. Neural Network (MLP)	Trains MLP (64→32 neurons), computes metrics	Accuracy, Precision, Recall, F1 printed + bar chart
13. Final Comparison	All three models compared on accuracy and F1	Two final bar charts
14. Advanced Heatmap	Seaborn correlation heatmap (2,000-row sample)	Color-coded heatmap
15. Interactive Plots	Plotly interactive class distribution + model accuracy	Interactive charts (open in browser)

8. Understanding the Outputs

Classification Metrics

After each model section, four metrics are printed:

- **Accuracy** — percentage of all predictions that were correct
- **Precision** — of all predicted attacks, how many were real attacks (avoids false alarms)
- **Recall** — of all real attacks, how many were caught (avoids missed detections)
- **F1 Score** — harmonic mean of precision and recall; the primary summary metric

Expected Results

Model	Accuracy	F1 Score
Logistic Regression	~0.8167	~0.7859
Random Forest	~0.9907	~0.9899
Neural Network (MLP)	~0.9751	~0.9743

Minor variations from these values are normal due to differences in the dataset version or system environment.

Computational Performance (qualitative)

Model	Training Speed	Memory
Logistic Regression	Fast	Low
Random Forest	Moderate	Moderate
Neural Network	Slow	High

9. Troubleshooting

FileNotFoundError: CICIOT23/train/train.csv

The dataset file is missing or in the wrong location. Re-read [Step 5](#) and verify the exact path.

ModuleNotFoundError: No module named 'sklearn' (or any other library)

Your virtual environment may not be active, or installation failed. Run:

```
bash  
  
pip install -r requirements.txt
```

Make sure `(venv)` is visible in your terminal prompt before running this.

UndefinedMetricWarning during model evaluation

This is expected. The CICIoT2023 dataset has severe class imbalance — some minority attack classes have very few samples, causing precision/recall to be undefined for those classes. The weighted average metrics are still valid.

Neural Network convergence warning: `max_iter=100` reached

This is intentional. The MLP is deliberately limited to 100 iterations to reflect edge-feasibility constraints and keep training time short for comparison purposes. To train to full convergence, change `max_iter=100` to `max_iter=500` in the MLP cell, but expect longer runtime.

Plotly charts do not display in the notebook

Plotly is configured to open charts in the browser (`pio.renderers.default = "browser"`). A new browser tab should open automatically. If it does not, change that line to:

```
python  
  
pio.renderers.default = "notebook"
```

Notebook runs very slowly

The Random Forest with 100 trees on 80,000 training samples can take several minutes on older hardware. To speed it up, reduce the sample size in the data loading cell:

```
python  
  
df = pd.read_csv(train_path, nrows=20000) # reduce from 100000
```

Note this will affect model accuracy.

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